High-Stakes Activation Probes on Indonesian

Ivan

Abstract

McKenzie *et al.* [9] trained linear probes on language model activations to detect high-stakes interactions. We test whether these probes generalize to a language absent from training by translating the evaluation datasets into Indonesian and evaluating without retraining. On Llama 3.1 8B (layer 12), mean AUROC on out-of-distribution benchmarks drops from 0.9046 to 0.8957 (0.9% degradation, 10-seed mean). On Gemma 3 12B (layer 32), the gap is 0.8972 to 0.8873 (1.0%). We then use Sparse Autoencoders (SAEs) to identify features enriched in the probe's error cases. On Llama, differential analysis identifies features with 10–15% higher prevalence in errors than in correct predictions, loosely aligned with technical-infrastructure and financial topics. On Gemma, the Gemma SAE produces no diagnostic signal. The cross-lingual degradation is small relative to the out-of-distribution gap between synthetic training data and naturalistic benchmarks (6–13%). The out-of-distribution gap is the more pressing concern for deployment.

1 Introduction

McKenzie *et al.* [9] introduced activation probes for detecting high-stakes LLM interactions—situations involving medical advice, legal counsel, financial decisions, or safety-critical scenarios. Their linear and attention probes, trained on residual stream activations, achieve strong performance on synthetic evaluation data but show degradation on naturalistic out-of-distribution benchmarks (Anthropic HH, ToolACE).

One practical question for deployment: do these probes work when users speak languages not represented in training data? If the probe relies on surface-level linguistic features, it would fail on new languages. If it captures a language-invariant concept in the model's representation space, it should transfer.

We test with Indonesian (Bahasa Indonesia), a language with approximately 200 million speakers that is typologically distant from the English/French/German/Hindi mix in the training data. Llama 3.1 8B [6] lists Indonesian as a Tier 2 supported language; Gemma 3 12B [4] includes Indonesian in its broader multilingual training. We evaluate on both models to check whether findings hold across architectures.

What we did. We translated all three evaluation sets to Indonesian, extracted activations from both models, and evaluated the trained probes without retraining. We then used Sparse Autoencoders—Llama Scope [7] and Gemma Scope 2 [5]—to investigate why probes fail on specific examples.

What we found.

- Cross-lingual degradation is small: 0.9% mean AUROC on Llama, 1.0% on Gemma.
- Earlier-to-middle layers generalise best on both models.

[0]**Work in progress.** Some further refinements and evaluations are still in progress. Thanks to Shivam Arora and the BlueDot Technical AI Safety Project Jan 2026 W4 group for feedback, and to BlueDot for facilitating and support.

Preprint. Under review.

- The synthetic-to-real domain gap is $\sim 10\times$ larger than the cross-lingual gap.
- Per-example agreement analysis reveals Gemma's failures are less consistent across languages (84.4% stable) than Llama's (91.7%).
- SAE failure analysis is modestly informative on Llama (10–15% differentials) but uninformative on Gemma ($<0.1\%$ differentials).

2 Setup

2.1 Methodology

We follow McKenzie *et al.* [9]:

- **Activations.** Residual stream at `input_layernorm` (before layer norm), mean-pooled over sequence length, with leading BOS token stripped. Max sequence length: 8,192 tokens.
- **Probe.** `StandardScaler + LogisticRegression` (L2, $C{=}10^{-3}$, `fit_intercept=False`, LBFGS solver, `max_iter=1000`).
- **Training data.** 8,000 examples (4,000 high-stakes, 4,000 low-stakes) from the paper's `prompts_4x` training set ($\approx$61% EN, $\approx$14% HI, $\approx$13% DE, $\approx$13% FR).
- **Seeds.** 10 random seeds (master seed 20250214); results are means with 95% bootstrap CIs. The L2-regularised LBFGS solver converges to the same optimum regardless of seed ($\sigma{=}0.000$ across all 10 seeds), so multi-seed serves as a sanity check rather than a variance estimate.
- **Models.** Llama 3.1 8B and Gemma 3 12B, loaded in 8-bit quantisation via bitsandbytes [3]. The paper uses fp16; we use 8-bit to fit on a single A10 GPU (24 GB).
- **Layer sweep.** Llama layers [12, 16, 20, 26, 31] of 32; Gemma layers [8, 16, 24, 31, 32, 41, 47] of 48.

We evaluate the linear probe only. The paper's best results use an attention probe on Llama-3.3-70B, which we did not run due to compute constraints.

2.2 Datasets

McKenzie *et al.* [9] generated the training and synthetic evaluation data using GPT-4o. Each example is a single-turn prompt labelled as high-stakes or low-stakes. The out-of-distribution evaluation sets differ structurally: Anthropic HH [1] consists of real multi-turn conversations; ToolACE [8] consists of multi-turn conversations with function/tool calls. Labels for both were assigned by GPT-4o on a 1–10 scale and binarised.

Table 1: Evaluation datasets used in this study.

Dataset	Orig size	ID size	Format
Synthetic test	2,000	1,950	Single-turn, multilingual, balanced
Anthropic HH	2,984	2,959	Multi-turn, real-world conversations
ToolACE	734	733	Multi-turn with function/tool calls

We refer to original-language datasets as "Original" rather than "English" because the training and synthetic test data are multilingual.

Indonesian translation. We translated all three evaluation sets into Indonesian using Claude Sonnet 4.5 (`claude-sonnet-4-5-20250929`) via the Anthropic API. The system prompt instructed the model to translate from any source language to Indonesian, prioritising native naturalness over literal accuracy, using dictionary-standard Indonesian (KBBI), preserving tone and formality, and keeping proper nouns, technical terms, and code unchanged. For multi-turn dialogues, each turn was translated independently while preserving role structure.

Translation quality was validated by the author (a native Indonesian speaker) across iterative prompt refinements. The Indonesian sets are slightly smaller because some examples were excluded due to translation API refusals.

2.3 Model Details

Table 2: Architecture details for the two evaluated models.

Property	Llama 3.1 8B	Gemma 3 12B
Layers	32	48
Hidden dim	4,096	3,840
ID support	Tier 2	Multilingual training
Quantisation	8-bit (bitsandbytes)	8-bit (bitsandbytes)

3 Results

3.1 Sanity Checks

Before interpreting results, we verify the probes learn a real signal.

Table 3: Sanity checks at the best probe layer for each model.

Check	Llama (L12)	Gemma (L32)
Training set AUROC	0.9997	0.9998
Random baseline AUROC	0.4441	0.4613
EN (train)	0.9980	0.9981
FR (train)	0.9995	0.9998
DE (train)	0.9999	1.0000
HI (train)	0.9978	0.9990

The per-language breakdown shows the probe performs uniformly across all four training languages on both models, consistent with a language-invariant representation being probed.

3.2 Layer Sweep

Table 4: Layer sweep for Llama 3.1 8B (32 layers). Mean AUROC across 10 seeds.

Layer	Orig Synth	Orig Anth	Orig Tool	ID Synth	ID Anth	ID Tool
12	**0.9985**	**0.9358**	**0.8734**	**0.9970**	**0.9044**	**0.8870**
16	0.9981	0.9077	0.8773	0.9964	0.8780	0.8707
20	0.9974	0.8992	0.8540	0.9952	0.8696	0.8607
26	0.9960	0.8649	0.8363	0.9910	0.8236	0.8543
31	0.9944	0.8570	0.8528	0.9852	0.7593	0.8417

Layer 12 is best for Llama (38% of network depth); layer 32 is best for Gemma (67%). On both models, performance on out-of-distribution benchmarks degrades at later layers while synthetic performance remains high across all layers. Figures 1 and 2 show the mean AUROC profiles.

3.3 Cross-Lingual Performance

At the best layer for each model, comparing Original and Indonesian on the out-of-distribution benchmarks:

For reference, the out-of-distribution gap *within* the same language is much larger. From Synthetic to Anthropic (Original): Llama drops 6.3%, Gemma drops 7.5%. From Synthetic to ToolACE (Original): Llama drops 12.5%, Gemma drops 12.8%.

Table 5: Layer sweep for Gemma 3 12B (48 layers). Mean AUROC across 10 seeds.

Layer	Orig Synth	Orig Anth	Orig Tool	ID Synth	ID Anth	ID Tool
8	0.9941	0.7872	0.7850	0.9898	0.7836	0.7838
16	0.9988	0.9065	0.8568	0.9977	0.8984	0.8665
24	0.9990	0.9342	0.8531	0.9982	0.9246	0.8389
31	0.9987	0.9209	0.8542	0.9976	0.9158	0.8452
32	**0.9987**	**0.9238**	**0.8706**	**0.9975**	**0.9176**	**0.8571**
41	0.9979	0.9055	0.8751	0.9953	0.8700	0.8594
47	0.9976	0.8832	0.8752	0.9943	0.8811	0.8515

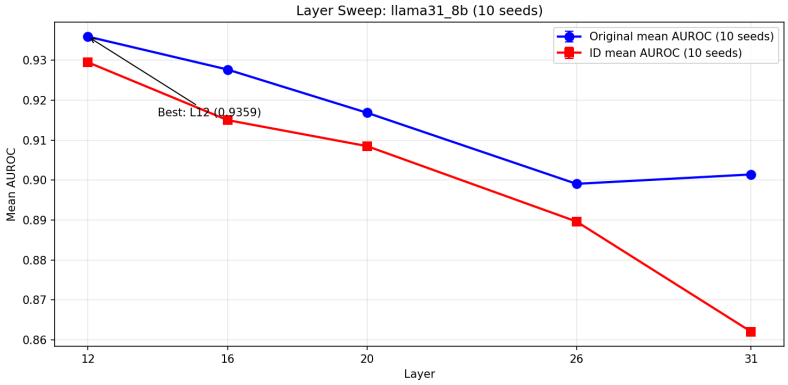


Figure 1: Layer sweep for Llama 3.1 8B. Mean AUROC across Synthetic, Anthropic HH, and ToolACE. Layer 12 is best for both Original (0.9359) and Indonesian (0.9295). The cross-lingual gap (vertical distance between curves) remains roughly constant across layers.

3.4 Cross-Lingual Agreement

Unlike AUROC, which measures aggregate discrimination, per-example agreement analysis reveals whether the probe fails on the *same* examples across languages. We compare predictions on Anthropic HH.

On Llama, 91.7% of predictions are stable across languages. On Gemma, only 84.4% are stable. Gemma's translation-induced failure rate (14.3%) is nearly three times Llama's (5.0%). This is notable because the aggregate AUROC gap is similar (1.0% vs. 0.9%): the headline numbers look comparable, but at the individual-example level Gemma has substantially more cases where the same input receives a different prediction depending on language.

3.5 Error Analysis

ToolACE errors are almost entirely false negatives. Both models produce near-zero false positives but miss 42–47% of high-stakes examples. ToolACE high-stakes scenarios involve tool/function calls where the stakes come from the *consequence* of the action, not from explicit danger language. The probe, trained on conversational data, does not recognise stakes in this format.

Anthropic HH is dominated by false negatives. FNs outnumber FPs by 10–12$\times$ on both models. Harmful intent is frequently embedded within a longer domestic or social narrative that distributes the signal across benign-sounding context. Mean-pooling over the full conversation pulls the pooled vector toward the low-stakes region when harmful content occupies only a small fraction of tokens.

Figure 4 shows that error examples in raw activation space are not geometrically distinct from correct predictions.

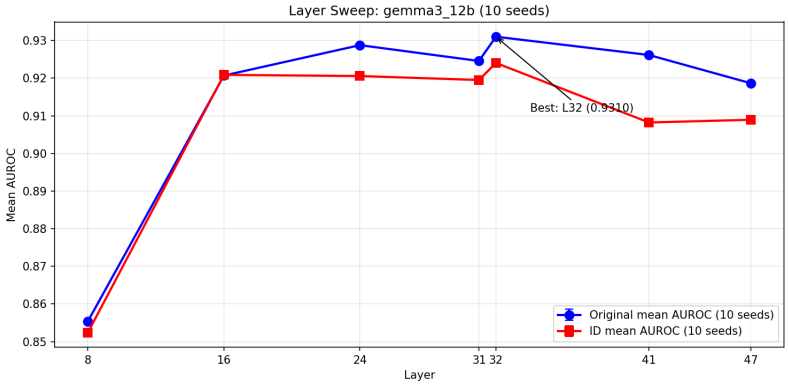


Figure 2: Layer sweep for Gemma 3 12B. Unlike Llama's monotonic decline, Gemma's probe quality rises through the first third of the network before plateauing and declining. Layer 32 is best for both Original (0.9310) and Indonesian (0.9241). The cross-lingual gap widens at deeper layers (41, 47).

Table 6: Cross-lingual AUROC at best layer. Delta = Indonesian − Original.

Model	Test Set	Original	Indonesian	Δ
Llama L12	Anthropic	0.9358	0.9044	−0.0314
	ToolACE	0.8734	0.8870	+0.0136
	Mean (OOD)	**0.9046**	**0.8957**	**−0.0089**
Gemma L32	Anthropic	0.9238	0.9176	−0.0062
	ToolACE	0.8706	0.8571	−0.0135
	Mean (OOD)	**0.8972**	**0.8873**	**−0.0099**

4 SAE Failure Analysis

4.1 Methodology

To investigate why probes fail, we decompose model activations using pre-trained Sparse Autoencoders (SAEs) loaded via SAELens [2]:

- **Llama.** Llama Scope [7] SAE at layer 12, residual position, 32K features (8× expansion). Model: `fnlp/Llama3_1-8B-Base-L12R-8x`.
- **Gemma.** Gemma Scope 2 [5] SAE at layer 41, 16K features. Model: `google/gemma-scope-2-12b-it`. The best probe layer is 32, but Gemma Scope 2 provides Neuronpedia dashboards [10] only at layers {12, 24, 31, 41}; we use layer 41.

The pipeline: (1) hook `input_layernorm` to capture per-token activations; (2) encode each token's activation through the SAE individually; (3) max-pool across the sequence to get one feature vector per example; (4) extract the top-10 features by activation strength; (5) compute differential prevalence between error and correct cases.

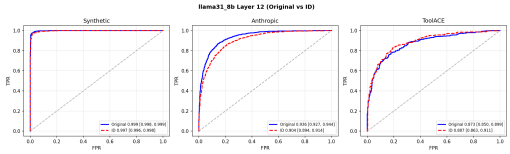
Why differential analysis? Naive top-k frequency is confounded by universally active features. Differential analysis (`error_rate` − `correct_rate`) controls for base rates, isolating features specifically enriched in errors.

4.2 Llama Results

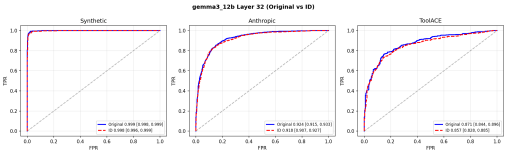
Sparsity. L0 ranged from ∼66 (Synthetic) to ∼205 (ToolACE) active features per example out of 32K.

Naive top-k (confounded). The most frequent features—604, 17722, 19230—appear in 100% of both errors and correct predictions. Neuronpedia labels these as generic conversational features.

Differential analysis. After controlling for base rates, some features are modestly enriched (Table 9).



(a) Llama 3.1 8B, layer 12. Blue = Original; red dashed = Indonesian.



(b) Gemma 3 12B, layer 32. Blue = Original; red dashed = Indonesian.

Figure 3: ROC curves on Synthetic, Anthropic HH, and ToolACE. The Synthetic curves are near-identical in both languages (≈0.999). On Anthropic, the Original curve sits visibly above Indonesian for Llama (0.936 vs. 0.904); the gap is smaller for Gemma (0.924 vs. 0.918). On ToolACE, Indonesian marginally exceeds Original on Llama (within bootstrap CIs).

Table 7: Per-example prediction agreement on Anthropic HH.

Behaviour	Llama (L12)	Gemma (L32)
Stable predictions	2713 (91.7%)	2499 (84.4%)
Both predicted high-stakes	2163 (73.1%)	1682 (56.8%)
Both predicted low-stakes	550 (18.6%)	817 (27.6%)
Translation-induced disagreements	246 (8.3%)	460 (15.6%)
High-stakes in Orig, low-stakes in ID	149 (5.0%)	423 (14.3%)
Low-stakes in Orig, high-stakes in ID	97 (3.3%)	37 (1.3%)

Technical-infrastructure conversations (deploying servers, managing databases) involve consequential actions but lack danger language, causing false negatives. Financial and historical topics sound serious to the probe but are often informational queries, causing false positives.

Signal-word analysis: we searched for trigger words ("emergency," "urgent," "danger," "kill," "hack") in false-positive texts. None appeared, indicating the probe's false positives at layer 12 are not driven by individual keywords.

Figure 5a confirms errors are not geometrically separable from correct predictions in raw activation space.

4.3 Gemma Results

Sparsity. L0 ≈ 1,614 active features per example out of 16K — an order of magnitude less sparse than Llama.

Differential analysis. All top-10 features appeared in errors and correct examples with <0.1% differential. The Neuronpedia labels are token-level fragments rather than semantic concepts (e.g., "pistachio, tomar, invertebr"). At L0 ≈ 1,614 (10% of all features active per example), the SAE does not decompose activations into the kind of sparse, interpretable concepts needed for this analysis.

4.4 Summary

5 Discussion

Out-of-distribution performance is the dominant concern. The cross-lingual gap (0.9–1.0% on OOD benchmarks) is small compared to the OOD gap itself. Within the original language, probe performance drops from >0.99 AUROC on synthetic data to 0.92–0.94 on Anthropic and 0.87–0.88 on ToolACE—a gap of 6–13%. Addressing distribution shift (multi-turn dialogue patterns, framing-buried harm, tool-use contexts) would yield more practical benefit than multilingual adaptation.

Why cross-lingual transfer works. By layer 12 of a multilingual model, the residual stream maps inputs from different languages into a shared semantic space. Wendler *et al.* [11] provide supporting evidence: in multilingual LLMs, middle-layer representations converge to a shared semantic space across languages, with language-specific encoding concentrated in the first and last layers. A linear

Table 8: Error counts at best layer, seed 0, threshold 0.5.

Model	Dataset	Errors	Rate	FP	FN
Llama L12	Synthetic	41 / 2000	2.1%	17	24
	Anthropic	651 / 2984	21.8%	51	600
	ToolACE	344 / 734	46.9%	0	344
Gemma L32	Synthetic	36 / 2000	1.8%	10	26
	Anthropic	589 / 2984	19.7%	91	498
	ToolACE	309 / 734	42.1%	1	308

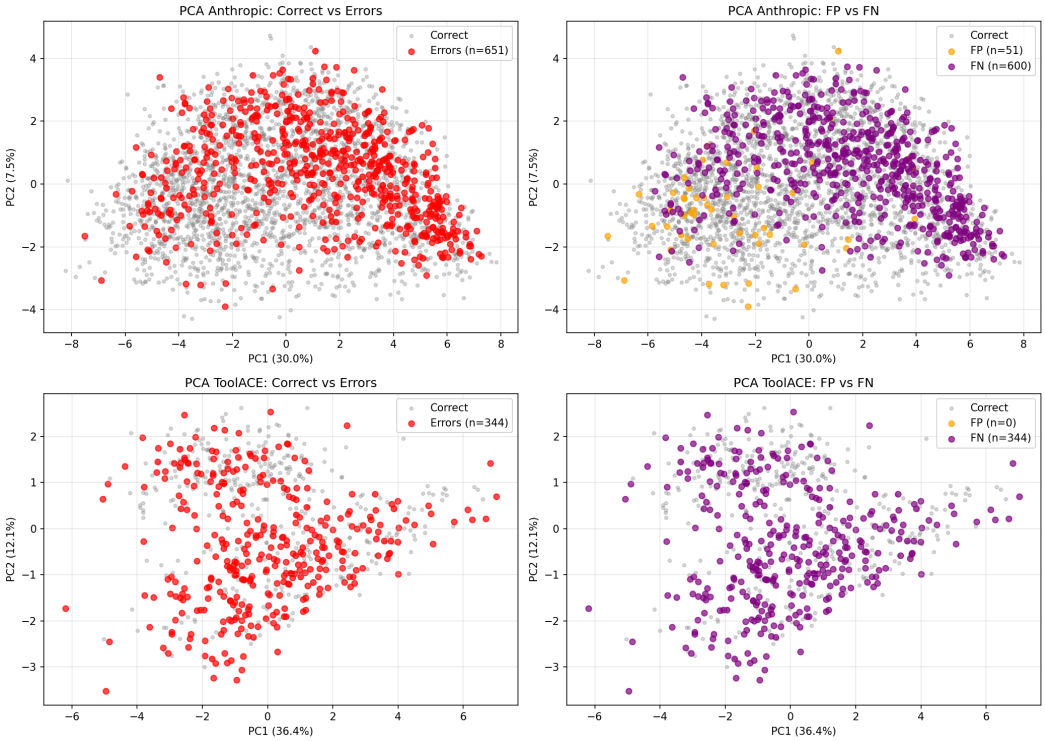


Figure 4: PCA of Llama 3.1 8B activations at layer 12 (Anthropic HH and ToolACE). Error examples (red) scatter throughout the same region as correct examples (grey). The ToolACE FP panel is empty—zero false positives on ToolACE.

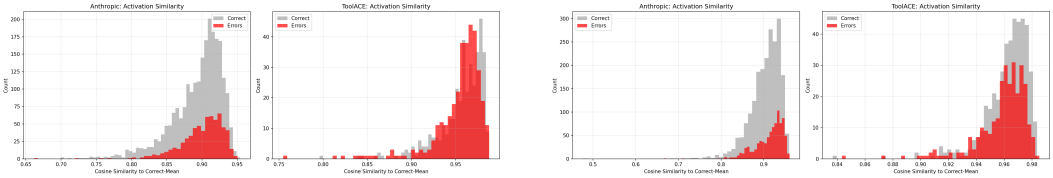
probe's decision boundary in this shared space separates "high-stakes" from "low-stakes" regardless of surface language.

Gemma's cross-lingual agreement is lower. The aggregate AUROC numbers for Llama and Gemma look similar, but the per-example agreement analysis tells a different story. Gemma has 14.3% of Anthropic examples where the probe is correct on the original language but fails on Indonesian, compared to Llama's 5.0%. For a deployed system, this per-example inconsistency matters: a user could receive correct safety classification in English but incorrect classification for the same query in Indonesian.

SAE analysis limitations. On Llama, differential feature analysis identified features enriched in errors with 10–15% differentials, but the features have broad semantic labels and do not pinpoint specific failure mechanisms. On Gemma, the analysis produced nothing. Two factors constrain this approach: (1) the available Gemma SAE at layer 41 has high L0 ($\approx$1,614), reducing discriminative power; (2) the Llama SAE is trained on the base model while we analyse the instruct model, introducing a distribution mismatch.

Table 9: Differentially enriched SAE features on Llama (layer 12, Anthropic HH). Differential = error rate − correct rate.

Feature	Error rate	Correct rate	Differential	Neuronpedia label
False-negative enriched				
6989	59.8%	44.4%	+15.3%	Technical / infrastructure concepts
16646	39.6%	25.4%	+14.3%	Knowledge and information
9766	23.0%	15.1%	+7.9%	Events and happenings
False-positive enriched				
2035	66.7%	45.0%	+21.7%	Financial statistics and reporting
6672	—	—	+13.2%	Historical references



(a) Llama 3.1 8B, layer 12.

(b) Gemma 3 12B, layer 41.

Figure 5: Cosine similarity of each example's activation to the mean correct-prediction activation. Errors (red) and correct predictions (grey) have nearly identical distributions on both models and datasets. Probe failures are not caused by a gross geometric mismatch in activation space.

6 Limitations

- **One language pair.** We test with Indonesian only. The cross-lingual gap perhaps could be larger for non-Latin-script languages (Arabic, Chinese, Thai) and also languages that has less speakers.
- **Translation quality.** Some performance differences may reflect translation artefacts. Translation was reviewed by a native speaker but not professional translators.
- **Linear probe only.** We did have only tested using a probe tested on mean pooled activation

7 Future Work

- **Alternative aggregation.** Other probes such as with Max-pooling, attention-weighted pooling, etc perhaps will show different results.
- **Iterative data generation.** The dominant error pattern—framing-buried harm—could be targeted by generating synthetic training examples that embed harmful intent within benign-sounding narratives.
- **Social-context ablation.** Stripping the surrounding domestic narrative from Anthropic HH false negatives and re-evaluating on the isolated harmful segment would test the mean-pooling dilution hypothesis.

8 Conclusion

On two models (Llama 3.1 8B and Gemma 3 12B), a multilingual-trained linear probe for high-stakes detection transfers to Indonesian with 0.9–1.0% mean AUROC degradation on out-of-distribution benchmarks. This cross-lingual gap is small relative to the out-of-distribution gap between synthetic training data and naturalistic benchmarks (6–13%).

Per-example agreement analysis shows Llama's probe predictions are more consistent across languages (92% stable on Anthropic HH) than Gemma's (84%). Gemma's higher translation-induced

Table 10: Comparison of SAE analyses.

Property	Llama (L12, 32K)	Gemma (L41, 16K)
L0 (sparsity)	66–205	≈1,614
Differential analysis	Modest (10–15%)	No signal (<0.1%)
Feature type	Semantic concepts	Token / subword patterns

failure rate (14.3% vs. 5.0%) indicates less reliable cross-lingual transfer at the individual-example level despite similar aggregate AUROC.

Probe failures concentrate in two patterns visible in both models: framing-buried harm (harmful intent embedded in domestic/social narratives, where mean-pooling dilutes the signal) and domain mismatch (tool-use formatted requests that the conversationally-trained probe does not recognise as high-stakes). An exploratory SAE-based analysis identified some features modestly enriched in error cases on Llama but produced no diagnostic signal on Gemma.

References

[1] Y. Bai, A. Jones, K. Ndousse, A. Askell, A. Chen, N. DasSarma, D. Drain, S. Fort, D. Ganguli, T. Henighan, N. Joseph, S. Kadavath, J. Kernion, T. Conerly, S. El-Showk, N. Elhage, Z. Hatfield-Dodds, D. Hernandez, T. Hume, S. Johnston, S. Kravec, L. Lovitt, N. Nanda, C. Olsson, D. Amodei, T. Brown, J. Clark, S. McCandlish, C. Olah, B. Mann, and J. Kaplan. Training a helpful and harmless assistant with reinforcement learning from human feedback, 2022. URL https://arxiv.org/abs/2204.05862.

[2] J. Bloom and SAELens Contributors. SAELens, 2024. URL https://github.com/jbloomAus/SAELens. GitHub repository.

[3] T. Dettmers, M. Lewis, Y. Belkada, and L. Zettlemoyer. LLM.int8(): 8-bit matrix multiplication for transformers at scale, 2022. URL https://arxiv.org/abs/2208.07339.

[4] Gemma Team. Gemma 3 technical report, 2025. URL https://arxiv.org/abs/2503.19786.

[5] Google DeepMind. Gemma Scope 2: Helping the AI safety community deepen understanding of complex language model behavior, 2025. URL https://deepmind.google/blog/gemma-scope-2-helping-the-ai-safety-community-deepen-understanding-of-complex-language-model-behavior/. Blog post.

[6] A. Grattafiori, A. Dubey, and et al. The Llama 3 herd of models, 2024. URL https://arxiv.org/abs/2407.21783.

[7] Z. He, W. Shu, X. Ge, L. Chen, J. Wang, Y. Zhou, F. Liu, Q. Guo, X. Huang, Z. Wu, Y.-G. Jiang, and X. Qiu. Llama Scope: Extracting millions of features from Llama-3.1-8B with sparse autoencoders, 2024. URL https://arxiv.org/abs/2410.20526.

[8] W. Liu, X. Huang, X. Zeng, X. Hao, S. Yu, D. Li, S. Wang, W. Gan, Z. Liu, Y. Yu, Z. Wang, Y. Wang, W. Ning, Y. Hou, B. Wang, C. Wu, X. Wang, Y. Liu, Y. Wang, D. Tang, D. Tu, L. Shang, X. Jiang, R. Tang, D. Lian, Q. Liu, and E. Chen. ToolACE: Winning the points of LLM function calling, 2024. URL https://arxiv.org/abs/2409.00920.

[9] A. McKenzie, U. Pawar, P. Blandfort, W. Bankes, D. Krueger, E. S. Lubana, and D. Krasheninnikov. Detecting high-stakes interactions with activation probes, 2025. URL https://arxiv.org/abs/2506.10805.

[10] Neuronpedia. Neuronpedia, 2024. URL https://neuronpedia.org. Web platform.

[11] C. Wendler, V. Veselovsky, G. Monea, and R. West. Do Llamas work in English? on the latent language of multilingual transformers, 2024. URL https://arxiv.org/abs/2402.10588.

A Per-Dataset Layer Sweep

B Reproducibility Details

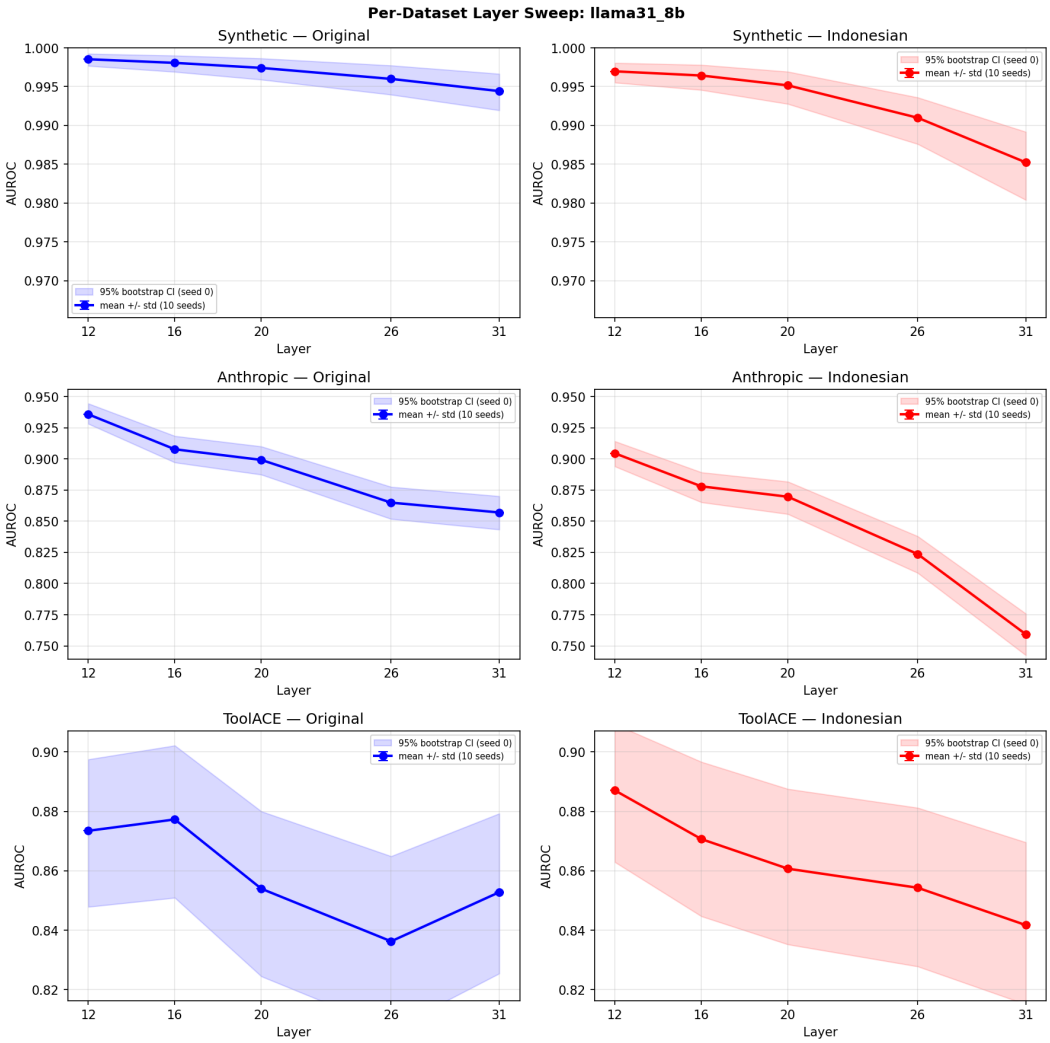


Figure 6: Per-dataset layer sweep for Llama 3.1 8B. Left column: Original. Right column: Indonesian. Shaded region is 95% bootstrap CI (10 seeds). Synthetic performance is high and flat across all layers. Anthropic performance degrades monotonically from layer 12. ToolACE shows high variance due to the small test size (734 examples).

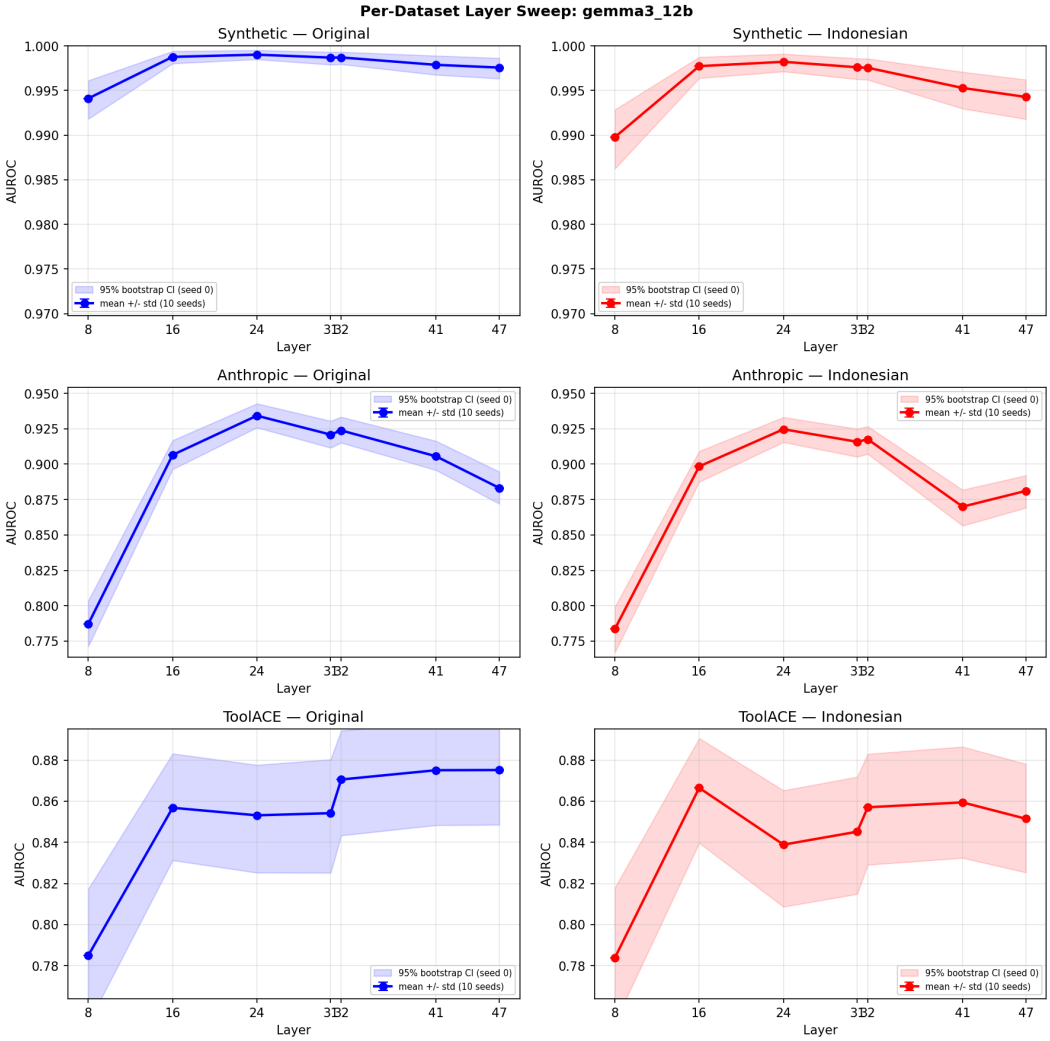


Figure 7: Per-dataset layer sweep for Gemma 3 12B. Left column: Original. Right column: Indonesian. Compared to Llama, Gemma shows a more gradual rise on Anthropic before peaking around layer 32. The Indonesian Anthropic curve shows a larger drop at deeper layers (41, 47) than the Original, reflecting the per-example inconsistency in Section 3.4.

Table 11: Full reproducibility details.

Parameter	Llama 3.1 8B	Gemma 3 12B
Model ID	`meta-llama/Llama-3.1-8B-Instruct`	`google/gemma-3-12b-it`
Quantisation	8-bit (bitsandbytes)	8-bit (bitsandbytes)
Best probe layer	12	32
Hidden dim	4,096	3,840
Activation hook	`model.model.layers[L] .input_layernorm`	`model.language_model .layers[L].input_layernorm`
Pooling	Mean pool over sequence, attention mask	
Max seq. length	8,192 tokens	
BOS handling	Strip leading BOS token	
Probe	`StandardScaler + LogisticRegression(C=1e-3, solver=lbfgs, max_iter=1000, fit_intercept=False)`	
Training set	8,000 examples (`prompts_4x/train.jsonl`)	
Seeds	10 (master seed 20250214)	
SAE (failure)	`fnlp/Llama-Scope` L12R-8x (32K)	`google/gemma-scope-2-12b-it` L41 (16K)
Translation model	`claude-sonnet-4-5-20250929`	
Compute	Lambda Labs A10 GPU (24 GB)	